

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.3 TAXONOMY II: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: How can we identify and classify living organisms accurately? You have studied the maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat from around your school. Now you will use everything you learned over 8 lessons to write a Final Explanation and to build your group's 'Local Biodiversity Classification Guide'. Answer each section in your own words, using evidence from your observations, keys and models. Use scientific words correctly (e.g. division, phylum, class, dichotomous key, virus). Write 4 to 6 sentences per section, and include at least one Kenyan example in each. Read the exemplar after you finish to compare, not to copy.

1. Why we need a standard system of classification

At the start, different groups sorted the school organisms by size, colour, habitat or usefulness as food, and the groupings did not agree. Explain why those groupings did not work for everyone and why scientists use shared, observable characteristics instead. Use at least two organisms from the school collection.

When we first sorted the organisms, one group put the maize plant, bean plant and goat together because they are food or feed. Another group put the mosquito, butterfly and chicken together because they can fly. Both groupings made sense to the group, but they were different, so a person from Kisumu and a person from Nairobi could not use them to find the same organism. Size, colour and usefulness change from place to place and from person to person. Scientists instead classify organisms by shared, observable and structural characteristics, such as having true roots and flowers, having a backbone, or the type of body covering. These features can be checked by anyone, anywhere, so the system is the same for everybody. This lets us organise biodiversity, identify organisms and communicate about them accurately.

2. How plants are classified

Explain how the plants in our collection (moss, fern, maize, bean plant, grass) are grouped

Plants are grouped into divisions by their structure and how they reproduce. Moss belongs to Bryophyta: it is small, has no true roots, stems or leaves, no vascular tissue, and grows in damp places such as shaded walls. The fern belongs to Pteridophyta: it has true roots, stems and leaves with vascular tissue, but reproduces using spores, usually found under the leaves. Maize, bean

into divisions. State the characteristic features you used for Bryophyta, Pteridophyta and Spermatophyta, and say which division each plant belongs to.	plant and grass belong to Spermatophyta, the seed plants, because they have vascular tissue and reproduce using seeds. Maize and grass are monocotyledons, with one seed leaf and parallel-veined leaves. The bean plant is a dicotyledon, with two seed leaves and net-veined leaves. By checking features like roots, vascular tissue, spores and seeds, we can place any plant, such as sukuma wiki or a Kericho tea bush, in the right group.
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3. How animals are classified

Explain how the animals in our collection (mosquito, butterfly, earthworm, snail, chicken, lizard, goat) are grouped into phyla and classes. Describe the features you used and name the group of at least four animals.	Animals are first placed into phyla by their body plan, then into classes by more detailed features. The earthworm has a long body divided into ring-like segments, so it belongs to Annelida. The snail has a soft body and a shell, so it belongs to Mollusca. The mosquito and butterfly have jointed legs, an exoskeleton and a body with three parts (head, thorax, abdomen) and six legs, so they belong to Arthropoda, class Insecta. The chicken, lizard and goat have a backbone, so they belong to Chordata. Within Chordata, the chicken has feathers and is in class Aves, the lizard has dry scaly skin and lays eggs with leathery shells, so it is Reptilia, and the goat has hair and feeds its young on milk, so it is Mammalia. Using these features, we can see that a goat is more closely related to a human than to a snail.
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4. Using a dichotomous key to identify an organism

Write a short dichotomous key (at least 3 steps) for either plants or animals from our school collection. Then explain how someone would use it to identify an unknown organism, and why each step must offer two contrasting choices.	Animal key: 1a. Body has a backbone – go to 2. 1b. No backbone – go to 3. 2a. Body covered with feathers – chicken. 2b. Body not covered with feathers – go to 4. 3a. Body has a shell – snail. 3b. Body has no shell – go to 5. 4a. Dry scales on skin – lizard. 4b. Hair on skin – goat. 5a. Body long and segmented, no legs – earthworm. 5b. Body has six jointed legs – insect. To use the key, you start at step 1, choose the statement that matches the organism, and follow the number until you reach a name. Each step gives two contrasting choices based on features you can see, so only one fits and you cannot be confused. If I find an unknown animal with no backbone, no shell, and six jointed legs, I follow 1b, 3b, 5b and identify it as an insect, like a butterfly. A good key uses clear features, not size or colour, which can vary.
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5. Viruses and why classification matters

Explain how viruses are different from cellular organisms and why they are important in nature. Then answer the driving question: how can we identify and classify living organisms accurately? Use a Kenyan example.	Viruses are very different from the organisms in our collection. They are not made of cells, they are much smaller than bacteria, and they can only reproduce inside the living cells of a host. Because of this, scientists do not place them easily in the plant or animal groups. Viruses are important in nature because of how they interact with living things. Some cause diseases, such as maize streak virus that damages maize crops in the Rift Valley, and HIV and influenza in people. Others, such as bacteriophages, infect bacteria and help control bacterial populations, and viruses are also used to make vaccines. To answer the driving question: we can identify and classify organisms accurately by observing shared structural features, grouping them into divisions, phyla and classes, and using dichotomous keys to identify unknown organisms. This standard system lets people everywhere, from Mombasa to Lake Victoria, study and talk about biodiversity in the same way.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of classification	Correctly places all organisms in the right divisions, phyla and classes and names accurate features such as vascular tissue, spores, seeds, backbone and body covering.	Places most organisms correctly and names some accurate features, with one or two small errors.	Places few organisms correctly or uses features that are not scientific, such as size, colour or usefulness.
Use of evidence from the school organisms	Uses specific observations from the collection (e.g. fern spores, earthworm segments, chicken feathers) to support every claim.	Uses some observations from the collection to support claims.	Makes general statements with little or no reference to the organisms observed.
Dichotomous key quality	Key has at least 3 steps with clear, contrasting, observable choices that lead to the correct name for every organism, and the student explains how to use it.	Key works for most organisms, with mostly contrasting choices, but the explanation of use is limited.	Key has unclear, non-contrasting or non-observable choices and does not lead reliably to an identification.
Explanation of viruses and the driving question	Clearly explains how viruses differ from cellular organisms, gives importance with Kenyan examples, and fully answers the driving question.	Explains some differences and one importance of viruses and gives a partial answer to the driving question.	Gives unclear or incorrect information about viruses and does not connect to the driving question.
Communication and Kenyan context	Writing is clear and well organised, uses scientific vocabulary correctly, and includes relevant Kenyan examples in each section.	Writing is mostly clear, uses some scientific vocabulary, and includes some Kenyan examples.	Writing is hard to follow, uses little scientific vocabulary and has few or no local examples.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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